



Java Programming

Chapter 4

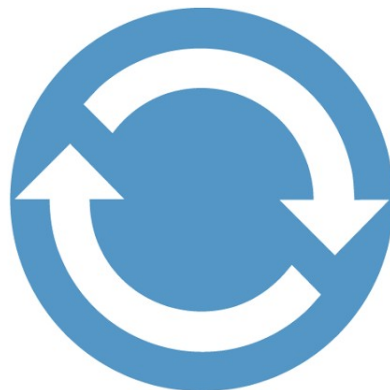
Loops

Contents

- What is loop?
- while
- do ~ while
- for

Loops is...

- Loops can execute a block of code as long as a specified condition is true.
- Loops are handy because they save time, reduce errors, and they make code more readable.
- Loops are needed in programming and work processes to repeat a set of actions automatically instead of writing or doing the same task over and over again by hand. They save time, make code short, and handle large amounts of data easily



Loops is...

```
doSomething();  
doSomething();  
doSomething();
```



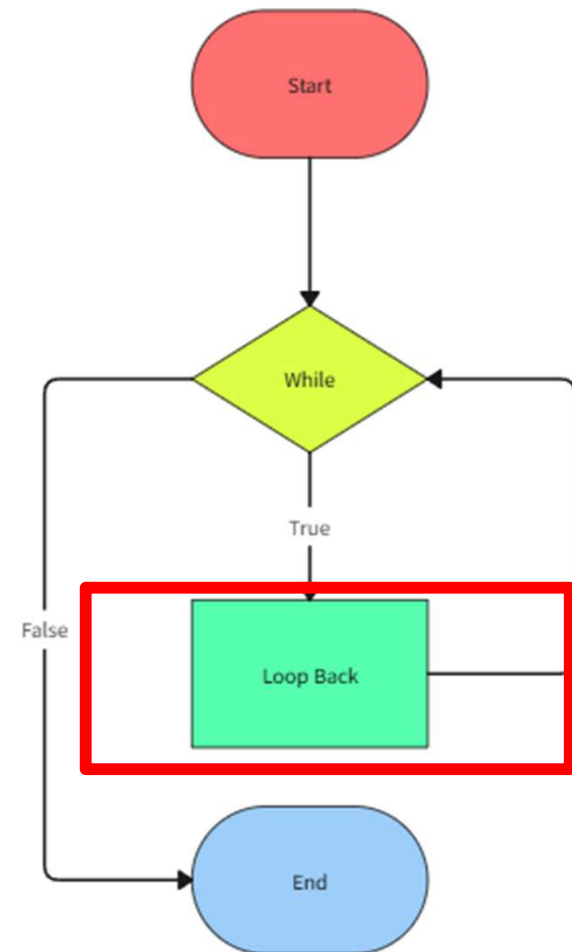
```
for (int i = 1; i <= 3; i++) {  
    doSomething();  
}
```

- Java supports loops
 - while()
 - do ~ while()
 - for()

while()

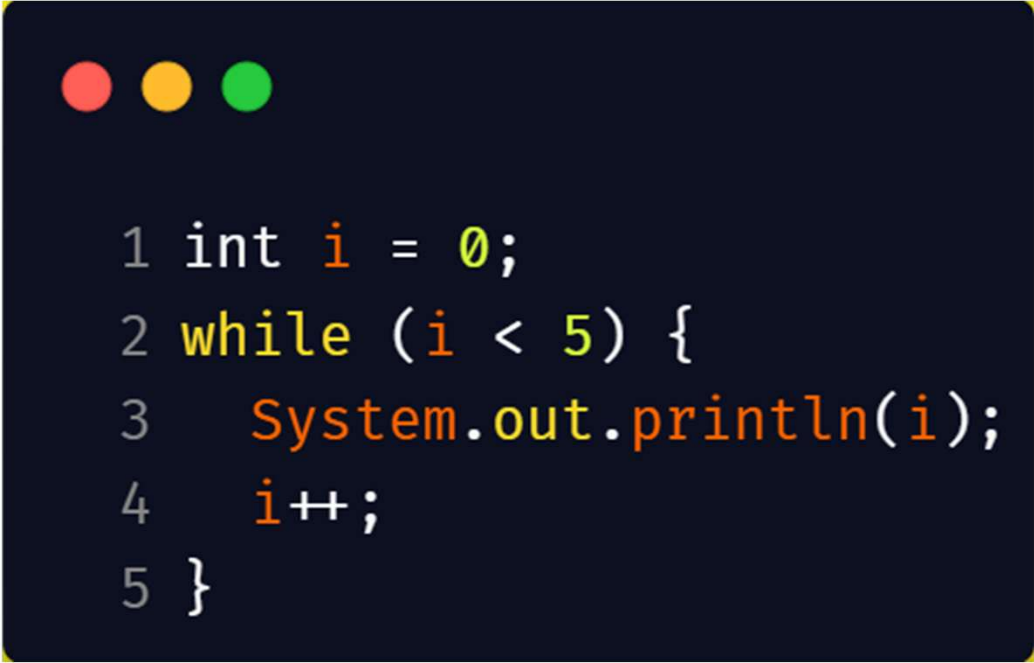
- Loops can execute a block of code as long as a specified condition is true.
- Loops are handy because they save time, reduce errors, and they make code more readable.

```
1 while (condition) {  
2   // code block to be executed  
3 }
```



while()

- In the example below, the code in the loop will run again and again, as long as a variable (i) is less than 5:
- Until $i < 5$ operation is true



```
1 int i = 0;
2 while (i < 5) {
3     System.out.println(i);
4     i++;
5 }
```

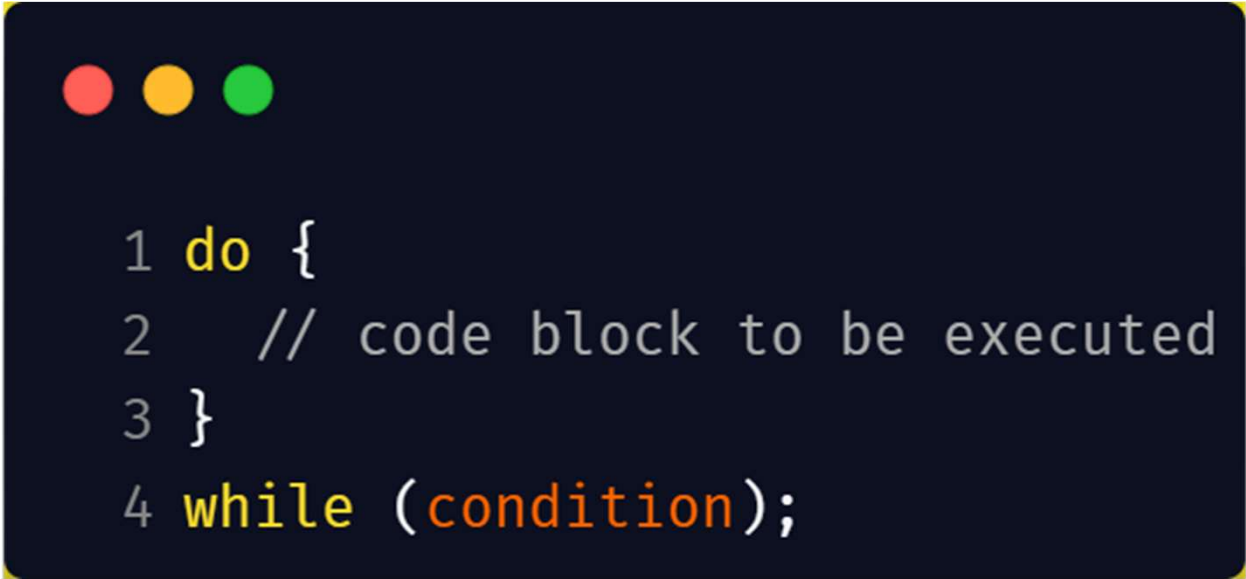
while()

- Countdown Example
 - You can also use a while loop to count down.
 - This example counts from 3 to 1, and then prints "Happy New Year!!" at the end:

```
1 int countdown = 3;
2
3 while (countdown > 0) {
4     System.out.println(countdown);
5     countdown--;
6 }
7
8 System.out.println("Happy New Year!! ");
```

The Do/While Loop

- The do/while loop is a variant of the while loop.
- This loop will execute the code block once, before checking if the condition is true. Then it will repeat the loop as long as the condition is true.
- Syntax

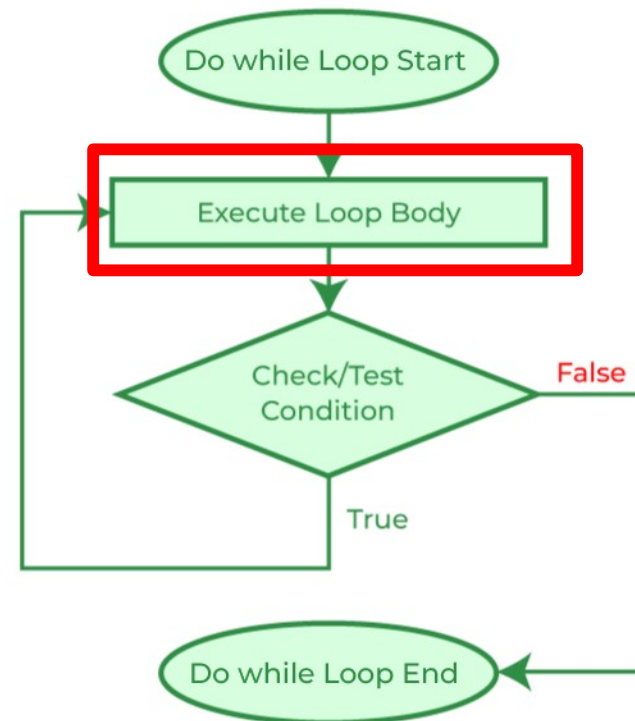


```
1 do {  
2     // code block to be executed  
3 }  
4 while (condition);
```

The Do/While Loop

- The example below uses a do/while loop.
- The loop will always be executed at least once, even if the condition is false, because the code block is executed before the condition is tested:

```
1 int i = 0;
2 do {
3     System.out.println(i);
4     i++;
5 }
6 while (i < 5);
```



The Do/While Loop

- Condition is False from the Start
 - In the while loop , we saw that if the condition is false at the beginning, the loop never runs at all.
 - The do/while loop is different: it will always run the code block at least once, even if the condition is false from the start.
 - *It executes at least once, even if the condition is not true.*

```
1 int i = 10;  
2  
3 do {  
4     System.out.println("i is " + i);  
5     i++;  
6 } while (i < 5);
```

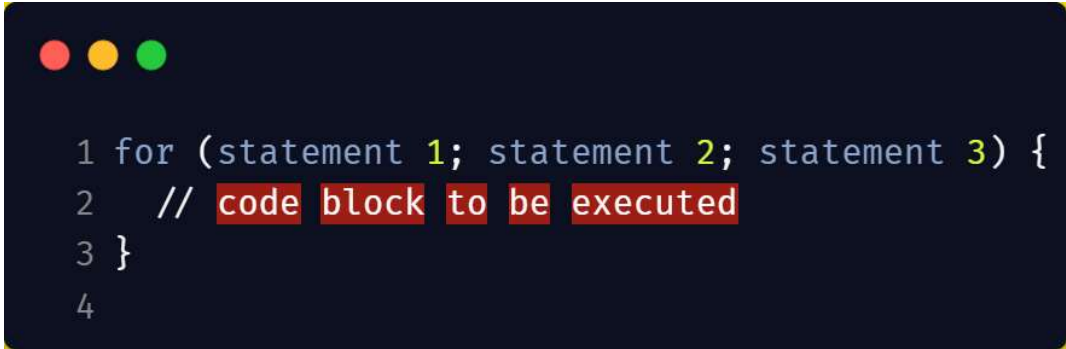
Excercise

- Let's find the sum of even numbers from 1 to 100.
- fill the white blank !!

```
1 int i = 
2 int sum = 
3
4 while ( ) {
5     if (  )
6         sum += i;
7     i++
8 }
9 System.out.println("from 1 to 100 even Sum = " + sum);
10 i++;
11 } while (i < 5);
```

For loop

- When you know exactly how many times you want to loop through a block of code, use the for loop instead of a while loop:
- Syntax

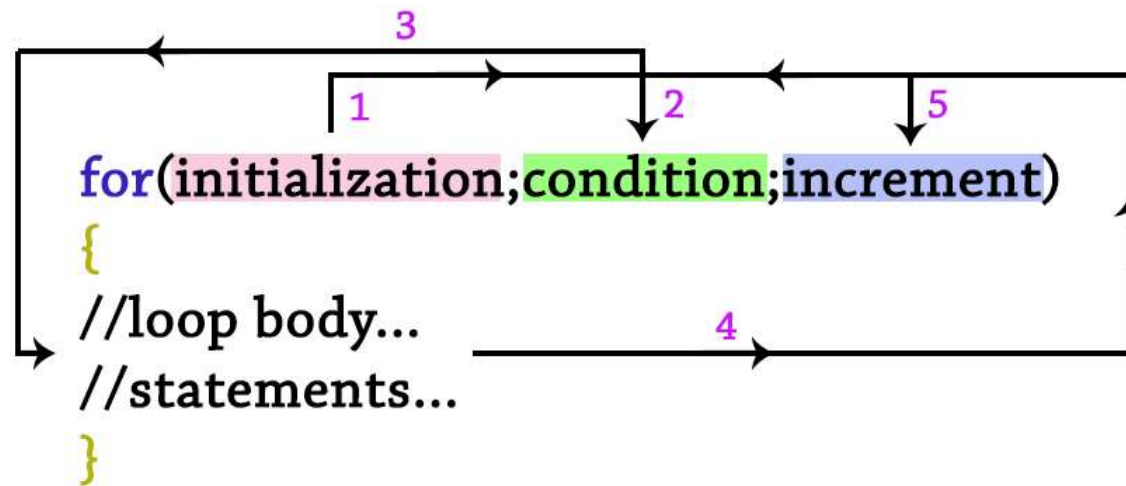


```
1 for (statement 1; statement 2; statement 3) {  
2     // code block to be executed  
3 }  
4
```

- Statement 1 is executed (one time) before the execution of the code block.
- Statement 2 defines the condition for executing the code block.
- Statement 3 is executed (every time) after the code block has been executed.

For loop

For Loop in Java



For loop

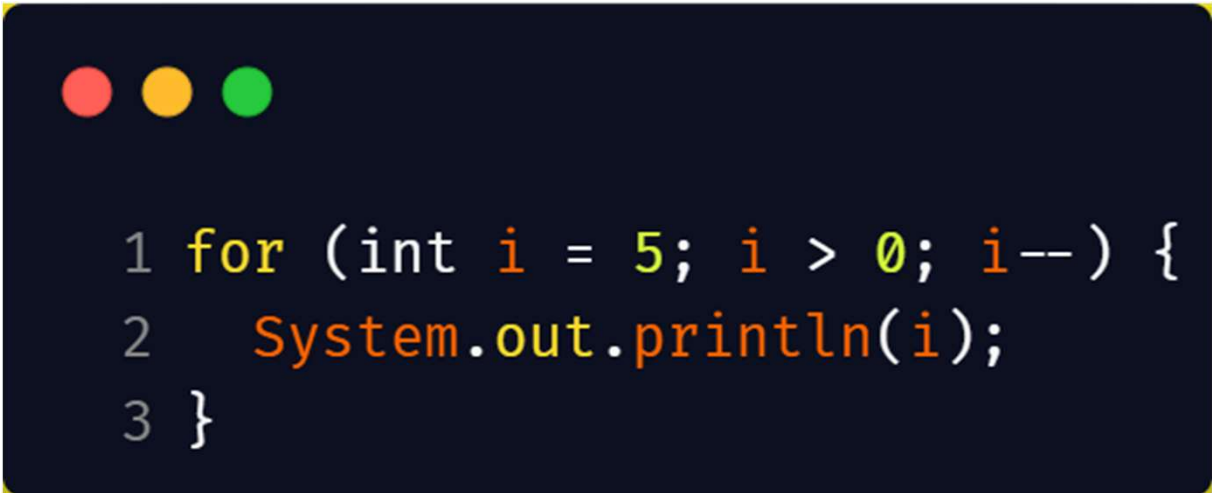
- Print Numbers
 - The example below will print the numbers 0 to 4:

```
1 int i = 0;
2 do {
3     System.out.println(i);
4     i++;
5 }
6 while (i < 5);
```

```
1 for (int i = 0; i < 5; i++) {
2     System.out.println(i);
3 }
```

For loop

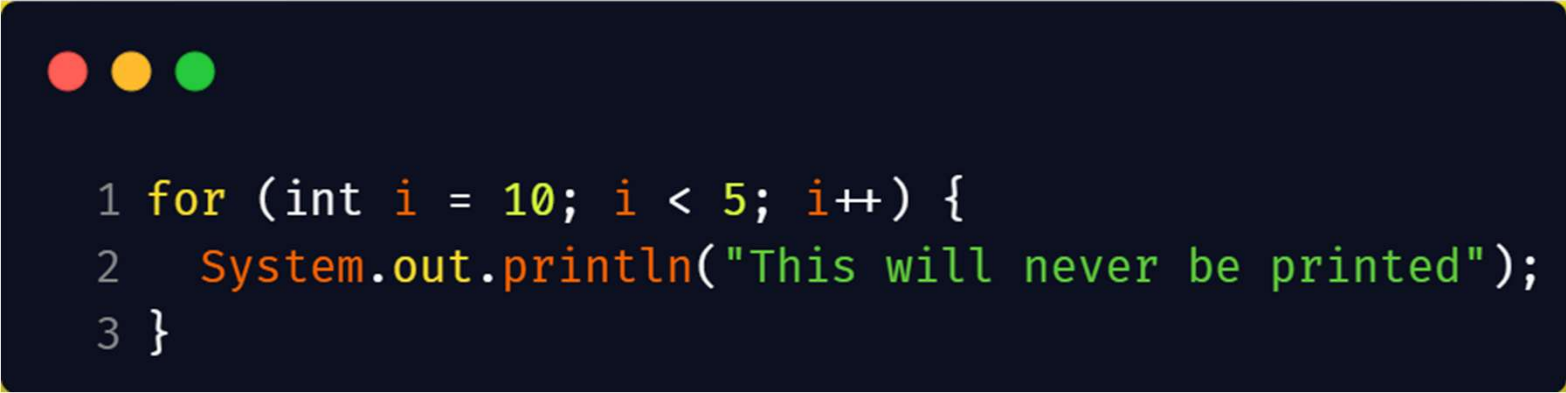
- Countdown



```
1 for (int i = 5; i > 0; i--) {  
2     System.out.println(i);  
3 }
```

For loop

- Why is there no output?



```
1 for (int i = 10; i < 5; i++) {  
2     System.out.println("This will never be printed");  
3 }
```

- Let's try to fix it

Nested Loops

- It is also possible to place a loop inside another loop. This is called a nested loop.
- The "inner loop" will be executed one time for each iteration of the "outer loop":

```
1 // Outer loop
2 for (int i = 1; i ≤ 2; i++) {
3     System.out.println("Outer: " + i); // Executes 2 times
4
5     // Inner loop
6     for (int j = 1; j ≤ 3; j++) {
7         System.out.println(" Inner: " + j); // Executes 6 times (2 * 3)
8     }
9 }
```

Nested Loops

- Print this result
- Fill in the blank

```
1 2 3
2 4 6
3 6 9
```

```
1 for (int i = ) {
2   for (int j = ) {
3     System.out.print(i * j + " ");
4   }
5   System.out.println();
6 }
7 }
```

Nested Loops

- prints the multiplication table for a specified number:

```
1 int number = 2;
2
3 // Print the multiplication table for the number 2
4 for (int i = 1; i ≤ 10; i++) {
5     System.out.println(number + " x " + i + " = " + (number * i));
6 }
```